HAT-P-36b

R-0960

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_230208` · created 2026-07-29T15:51:37+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459983.9199 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.019
Transit depth (Rp/Rs)^2	2.47 +/- 0.6 [%]
Orbital Inclination (inc)	88.55 +/- 2.59
Airmass coefficient 1 (a1)	1.018 +/- 0.015
Airmass coefficient 2 (a2)	-0.017 +/- 0.055
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#3 - [98, 45]
Optimal Method	PSF photometry
Transit Duration (day)	0.0617 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.019 vs 0.126 (deviation 1.63σ)
Duration fit vs published	0.0617 d vs 0.0925 d (deviation 7.33σ)
Mid-time O–C	0.6 min
Depth SNR	8.3
Residual scatter	1.42 %

Light curve

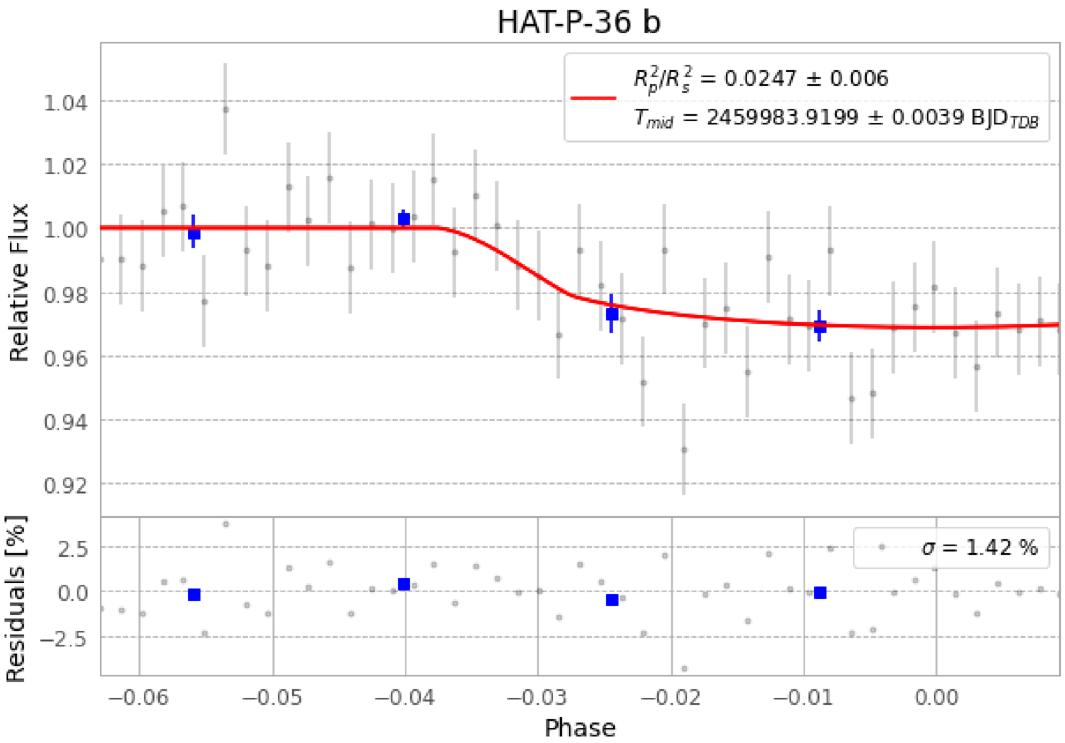

Figure 1 — FinalLightCurve_HAT-P-36 b_2023-02-08.png (SHA-256 6a84be44c1878871…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [558, 129], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 aeeef9a5e474fe8e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

47 FITS frames (31 MB), `HATP-36230208075715.FITS` → `HATP-36230208101515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230208.log (b5a7bd062a5a…), FinalLightCurve_HAT-P-36 b_2023-02-08.png (6a84be44c187…), FinalLightCurve_HAT-P-36 b_2023-02-08.csv (af908207d40f…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)